

Quiz 1 (Version 3)

CAS CS 132: *Geometric Algorithms*

February 2, 2025

Name: *Nathan Mull*

BUID: *12345678*

- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Row Operations

Apply the row operations:

$$R_2 \leftarrow -4R_2$$

$$R_2 \leftrightarrow R_3$$

$$R_2 \leftarrow R_2 + R_3$$

from top to bottom to the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} -2 & 6 & -10 \\ -7 & -7 & 0 \\ -8 & -5 & -2 \end{bmatrix} \xrightarrow{R_2 \leftarrow -4R_2}$$

$$\begin{bmatrix} -2 & 6 & -10 \\ 28 & 28 & 0 \\ -8 & -5 & -2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3}$$

$$\begin{bmatrix} -2 & 6 & -10 \\ -8 & -5 & -2 \\ 28 & 28 & 0 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_3}$$

Solution.

$$\begin{bmatrix} -2 & 6 & -10 \\ 20 & 23 & -2 \\ 28 & 28 & 0 \end{bmatrix}$$

2 Reduced Echelon Forms

Determine the reduced echelon form of the following matrix. You must write down the intermediate matrices and row operations you used in your calculation.

$$\begin{bmatrix} -1 & -2 & 3 & -14 \\ 1 & 2 & 0 & -4 \\ 3 & 6 & -10 & 48 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1}$$

$$\begin{bmatrix} -1 & -2 & 3 & -14 \\ 0 & 0 & 3 & -18 \\ 3 & 6 & -10 & 48 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{bmatrix} -1 & -2 & 3 & -14 \\ 0 & 0 & 3 & -18 \\ 0 & 0 & -1 & 6 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3R_3} \begin{bmatrix} -1 & -2 & 3 & -14 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 6 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 + 3R_3} \begin{bmatrix} -1 & -2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 6 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow -R_3} \begin{bmatrix} -1 & -2 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -6 \end{bmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{bmatrix} 1 & 2 & 0 & -4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -6 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3}$$

Solution.

$$\begin{bmatrix} 1 & 2 & 0 & -4 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3 General Form Solutions

Determine a general form solution for a linear system whose augmented matrix is row equivalent to the following matrix (*Hint*. First determine if the following matrix is in reduced echelon form).

$$\begin{bmatrix} 1 & 0 & 1 & -4 & 1 & 0 \\ 0 & 1 & 2 & -5 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -4 & 0 & -1 \\ 0 & 1 & 2 & -5 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$x_1 = -1 - 4x_3 + 4x_4$$

$$x_2 = -4 - 2x_3 + 5x_4$$

x_3 is free

x_4 is free

$$x_5 = 1$$